
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Asaba, Joshua
Date of birth:	2/24/1990
Subject:	030-Flight Performance&Planing
Result:	28,6 %
Time used:	00:00:13 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	379	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
2	361	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
3	420	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
4	381	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
5	377	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
6	391	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
7	384	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
8	382	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
9	374	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
10	390	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
11	378	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
12	315	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	1,00	1,00
13	373	31.	MASS AND BALANCE - AEROPLANES/HELICOPTERS	0,00	1,00
14	257	31.3.1	Definition of centre of gravity	0,00	1,00
15	278	31.3.3	Basic calculations of CG	0,00	2,00
16	424	32.	PERFORMANCE - AEROPLANES	0,00	1,00
17	328	32.	PERFORMANCE - AEROPLANES	0,00	1,00
18	77	32.2.5.4	Landing	1,00	1,00
19	152	32.2.5.4	Landing	1,00	1,00
20	231	32.2.5.4	Landing	0,00	1,00
21	225	32.3.3.1	Take-off	1,00	1,00
22	31	32.4.1	Take - off	0,00	1,00
23	135	32.4.1.7	Take-off climb	0,00	1,00
24	146	32.4.1.7	Take-off climb	0,00	1,00
25	98	32.4.1.7	Take-off climb	0,00	1,00
26	202	32.4.3	Cruise	1,00	1,00
27	15	32.4.3	Cruise	1,00	1,00
28	47	32.4.3	Cruise	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	96	32.4.3.5	Influence of variables on cruise performance	1,00	1,00
30	196	32.4.5	Descent	0,00	1,00
31	418	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
32	415	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
33	414	33.	FLIGHT PLANNING AND FLIGHT MONITORING	0,00	1,00
34	413	33.	FLIGHT PLANNING AND FLIGHT MONITORING	1,00	1,00
35	417	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	0,00	1,00
36	419	33.1	FLIGHT PLANNING FOR VFR FLIGHTS	0,00	1,00
37	259	33.1.1.5	Completion of navigation plan	0,00	2,00
38	175	33.2.1.2	Courses and distances from en-route charts	0,00	1,00
39	122	33.3.3	Specific fuel calculation procedures	0,00	1,00
40	30	33.5.1.1	Format of Flight Plan	0,00	1,00
Total:29%				12,00	42,00

Attachment 2: List of result by topic

Licence

Date: 18.07.2014

Examinee: Asaba, Joshua

Location: Uganda Civil Aviation Authority

Your Result: 12,00 from 42,00 Points (28,57 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
31. MASS AND BALANCE - AEROPLANES/HELICOPTERS	15	5,00	16,00	31,25
31.3. FUNDAMENTALS OF CG CALCULATIONS	2	0,00	3,00	0,00
32. PERFORMANCE - AEROPLANES	15	6,00	15,00	40,00
32.2. PERFORMANCE CLASS B - SINGLE-ENGINE AEROPLANES	3	2,00	3,00	66,67
32.3. PERFORMANCE CLASS B - MULTI-ENGINE AEROPLANES	1	1,00	1,00	100,00
32.4. PERFORMANCE CLASS A - AEROPLANES CERTIFICATED UNDER CS 25 ONLY	9	3,00	9,00	33,33
33. FLIGHT PLANNING AND FLIGHT MONITORING	10	1,00	11,00	9,09
33.1. FLIGHT PLANNING FOR VFR FLIGHTS	3	0,00	4,00	0,00
33.2. FLIGHT PLANNING FOR IFR FLIGHTS	1	0,00	1,00	0,00
33.3. FUEL PLANNING	1	0,00	1,00	0,00
33.5. ICAO FLIGHT PLAN (ATS Flight Plan)	1	0,00	1,00	0,00
Total	40	12,00	42,00	28,57

Candidate: Joshua Asaba
Date of birth: 1990-02-24

Subject: 030-Flight Performance&Planing
Date: 2014-07-18



1

How much weight must be shifted from Station 150.0 to Station 30.0 to move the CG to exactly the aft CG limit?

Total weight 7,500 lb

CG location Station 80.5

☐ 68.9 pounds.

☐ 62.5 pounds.

☒ 65.8 pounds.

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2

What is the maximum allowable weight that may be carried on a pallet which has the dimensions of 24.6 x 68.7 inches?
Floor load limit - 85 lb/sq ft
Pallet weight - 44 lb
Tiedown devices - 29 lb

☐ 968.6 pounds.

☐ 953.6 pounds.

☒ 924.5 pounds.

3

The center of gravity tube is

- ☐ shortened for lighter pilots.
- ☒ lengthened for heavier pilots.
- ☐ lengthened for lighter pilots.

4

Could 100 pounds of weight be shifted from Station 30.0 to Station 120.0 without exceeding the aft CG limit?

Total weight 4,750 lb

CG location Station 115.8

☐ No; the new CG would be located at Station 118.15.

☒ Yes; the new CG would be located at Station 117.69.

☐ Yes; the CG would remain at Station 115.8.

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5

What is the location of the CG if 146 pounds are removed from Station 150?

Aircraft weight 7,152 lb

CG location Station 82

☐ 83.4.

☒ 80.6.

☐ 81.3.

6

As the CG location is changed, recovery from a stall becomes progressively

- ☒ more difficult as the CG moves rearward.
- ☐ more difficult as the CG moves either forward or rearward.
- ☒ less difficult as the CG moves rearward.

7

An aircraft is loaded with the CG at the aft limit. What are the performance characteristics compared with the CG at the forward limit?

- ☒ **The aft CG provides the highest stall speed and cruising speed.**
- ☐ **Cruising speed is lower because of more induced drag created by the elevator or stabilizer being required to provide more lift with an aft CG.**
- ☐ **The aft CG provides the lowest stalling speed, the highest cruising speed, and least stability.**

8

The center of gravity of an aircraft can be determined by

- ☐ dividing total arm by total moments.
- ☐ multiplying total arm by total weight.
- ☒ dividing total moment by total weight.

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9

How much weight could be added at Station 120 without exceeding the aft CG limit?

Aircraft weight 9,500 lb

CG location Station 90.0

Aft CG limit Station 90.5

☐ 61.0 pounds.

☐ 161.0 pounds.

☒ 110.5 pounds.

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10

Under which condition is a forward CG most critical?

- ☐ On landing.
- ☒ On takeoff.
- ☐ When in an unusual attitude.

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11

What is the location of the CG if 60 pounds are removed from Station 70?

Aircraft weight 8,420 lb

CG location Station 85

☒ 84.1.

☐ 84.9.

☐ 85.1.

12

The term 'weigh-off' as used in ballooning means to determine the

- ☒ static equilibrium of the balloon as loaded for flight.
- ☐ standard weight and balance of the balloon.
- ☐ amount of fuel required for ascent to a preselected altitude.

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13

How much weight could be added at Station 160 without exceeding the aft CG limit?

Aircraft weight 8,300 lb

CG location Station 90.0

Aft CG limit Station 90.5

☐ 13.9 pounds.

☒ 16.5 pounds.

☐ 59.7 pounds.

14

When an aircraft is stationary on the ground, its total weight will act vertically

- ☐ *through its centre of pressure.*
- ☒ *through a point defined as the datum point.*
- ☐ *through the main wheels of its undercarriage assembly.*
- ☐ *through its centre of gravity.*

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15

(Please view annex 031-038 rev. 05.11.2004)
For the following boom to be in balance:

See Attachments:

1. 031-038.gif

Annex

(Attachment
No.1)

☒ $A = B (F_a + F_b)$

☐ $A = B \times F_b / F_a$

☐ $A = B \times F_a / F_b$

☐ $A = B + F_b / F_a$

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16

To level off from a descent maintaining the descending airspeed, the pilot should lead the desired altitude by approximately

☐ 50 feet.

☒ 60 feet.

☐ 20 feet.

17

(Refer to Figures 84 and 85.) What is the approximate fuel consumed when holding under Operating Conditions H-4?

☐ 2,550 pounds.

☒ 3,050 pounds.

☐ 3,190 pounds.

See Attachments:

1. atp_84.jpg	figure 84	(Attachment No.2)
2. atp_85.jpg	figure 85	(Attachment No.3)

18

A runway is contaminated with 0.5 cm of wet snow. The flight manual of a light twin nevertheless authorises a landing in these conditions. The landing distance will be, in relation to that for a dry runway:

☒ increased

☐ substantially decreased

☐ reduced

☐ unchanged

19

Which of the following factors will lead to an increase of ground distance during a glide, while maintaining the appropriate minimum glide angle speed?

- ☒ Tailwind.
- ☐ Decrease of aircraft mass.
- ☐ Headwind.
- ☐ Increase of aircraft mass.

20

To minimize the risk of hydroplaning during landing the pilot should:

- ☒ make a "positive" landing and apply maximum reverse thrust and brakes as quickly as possible.
- ☐ use normal landing-, braking- and reverse technique.
- ☐ postpone the landing until the risk of hydroplaning no longer exists.
- ☒ use maximum reverse thrust, and should start braking below the hydroplaning speed.

21

Take-off run is defined as the

- ☐ distance to V1 and stop, assuming an engine failure at V1.
- ☒ horizontal distance along the take-off path from the start of the take-off to a point equidistant between the point at which VLOF is reached and the point at which the aeroplane is 35 ft above the take-off surface.
- ☐ Distance from brake release to V2.
- ☐ distance to 35 feet with an engine failure at V1 or 115% all engine distance to 35 feet.

22

How is V2 affected if take off flaps 20° is chosen instead of take off flaps 10°?

- ☒ V2 increases in proportion to the angle at which the flaps are set.
- ☐ V2 decreases if not restricted by VMCA.
- ☐ V2 has the same value in both cases.
- ☐ V2 has no connection with T/O flap setting, as it is a function of runway length only.

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23

With an true airspeed of 194 kt and a vertical speed of 1 000 ft/min, the climb gradient is about :

☐ 3°

☒ 5°

☐ 0.08

☐ 0.03

24

If the take-off mass of an aeroplane is brake energy limited a higher uphill slope would

- ☐ have no effect on the maximum mass for take-off.
- ☒ increase the maximum mass for take-off.
- ☐ decrease the maximum mass for take-off.
- ☒ decrease the required take-off distance.

25

A higher outside air temperature

- ☐ does not have any noticeable effect on climb performance.
- ☐ reduces the angle and the rate of climb.
- ☒ increases the angle of climb but decreases the rate of climb.
- ☐ reduces the angle of climb but increases the rate of climb.

26

For a piston engined aeroplane, the speed for maximum range is :

- ☐ that which gives the minimum value of drag.
- ☐ 1.4 times the stall speed in clean configuration.
- ☐ that which gives the maximum value of lift
- ☒ that which gives the maximum lift to drag ratio.

27

The maximum horizontal speed occurs when:

- ☐ The thrust is equal to minimum drag.
- ☒ The maximum thrust is equal to the total drag.
- ☐ The thrust is equal to the maximum drag.
- ☐ The thrust does not increase further with increasing speed.

28

The optimum cruise altitude is

- ☐ the pressure altitude at which the best specific range can be achieved.
- ☒ the pressure altitude at which the speed for high speed buffet as TAS is a maximum.
- ☐ the pressure altitude up to which a cabin altitude of 8000 ft can be maintained.
- ☐ the pressure altitude at which the fuel flow is a maximum.

29

The Density Altitude

- ☒ is used to determine the aeroplane performance.
- ☐ is used to establish minimum clearance of 2.000 feet over mountains.
- ☐ is used to calculate the FL above the Transition Altitude.
- ☐ is equal to the pressure altitude.

30

A jet aeroplane descends with constant Mach number. Which of the following speed limits is most likely to be exceeded first?

- ☒ **Maximum Operating Speed**
- ☐ **High Speed Buffet Limit**
- ☐ **Never Exceed Speed**
- ☒ **Maximum Operational Mach Number**

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31

If an aircraft has a transponder, encoding altimeter, and DME, the proper equipment suffix to be entered on a flight plan is

☐ U.

☒ R.

☐ A.

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32

If fuel consumption is 15.3 gallons per hour and groundspeed is 167 knots, how much fuel is required for an aircraft to travel 620 NM?

☐ 60 gallons.

☒ 63 gallons.

☐ 57 gallons.

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33

GIVEN:

Usable fuel at takeoff 40 gal
Fuel consumption rate 12.2 gal/hr
Constant groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☐ 215 NM.

☒ 121 NM.

☐ 176 NM.

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34

GIVEN:

Usable fuel at takeoff 40 gal, Fuel consumption rate 12.2 gal/hr Constant
groundspeed 120 kts
Flight time since takeoff 1 hr 30 min

☒ 121 NM.

☐ 156 NM.

☐ 216 NM.

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35

How long will a Flight Service Station hold a VFR flight plan past the proposed departure time?

☒ **30 minutes.**

☐ **2 hours.**

☐ **1 hour.**

36

All fuel tanks should be fired during preflight to determine

- ☒ if there are any leaks in the tanks.
- ☐ if the pilot light functions properly on each tank.
- ☐ burner pressure and condition of the valves.

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37

Given: true course (TC) 017; W/V 340/30; TAS 420 kt
Find: wind correction angle (WCA) and ground speed (GS)

☐ WCA +2° ; GS 396 kt

☒ WCA +2° ; GS 416 kt

☐ WCA -2° ; GS 396 kt

☐ WCA -2° ; GS 426 kt

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38

You are flying a constant compass heading of 252° . Variation is 22°E , deviation is 3°W and your INS is showing a drift of 9° right. True track is ?

☐ 224°

☐ 280°

☒ 262°

☐ 242°

39

An aircraft flies at an airspeed of 380 kt. It flies from A to B and back to A. Distance AB = 480 NM. When going from A to B, it experiences a headwind component = 60 kt. The wind remains constant. The duration of the flight will be:

☒ 2h 10min

☐ 2h 32min

☐ 3h 00min

☐ 2h 35min

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40

If your destination airport has no ICAO indicator, in the appropriate box of your flight plan, you write:

☐ C ///

☒ B XXXX

☐ A AAAA

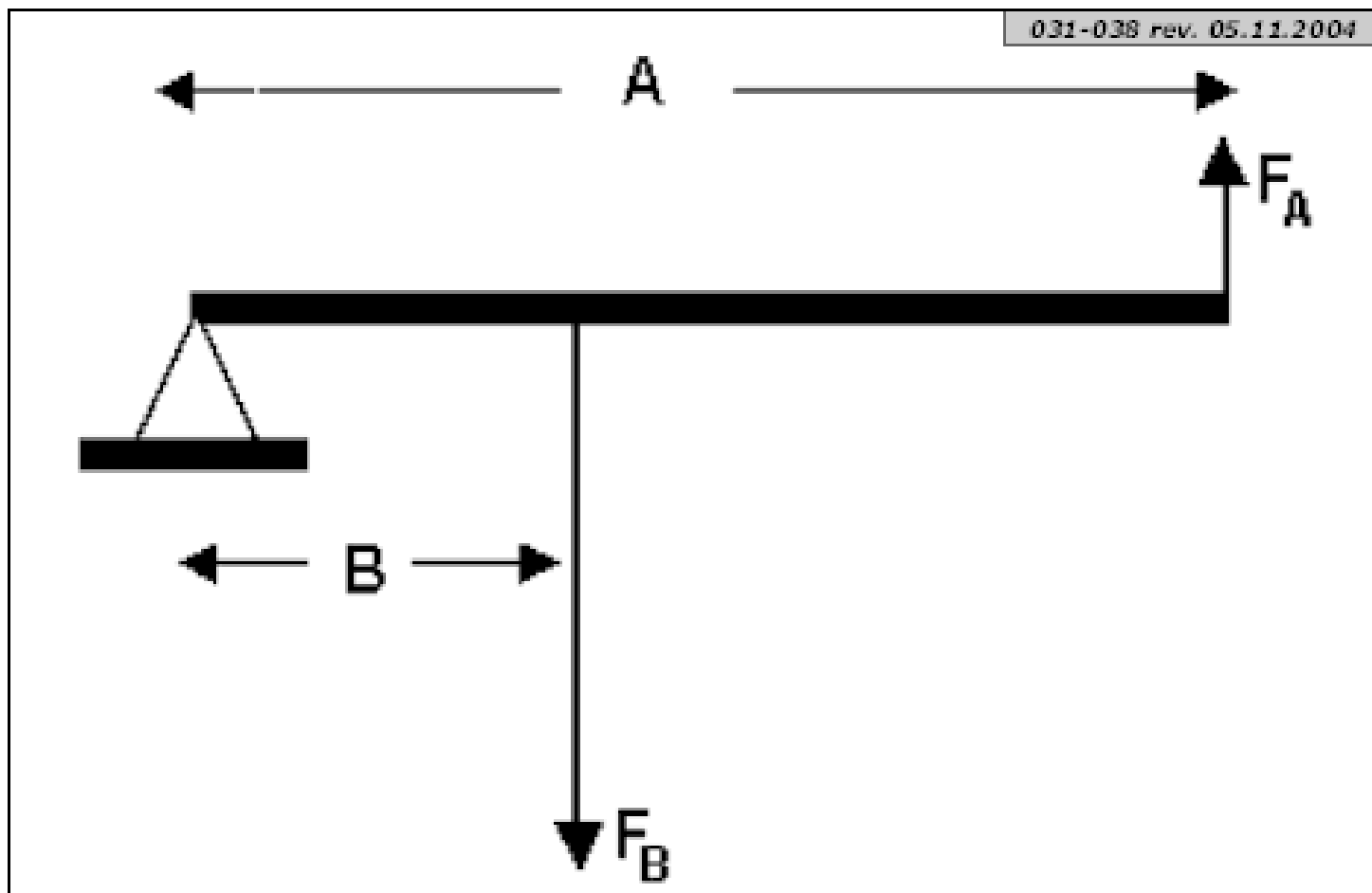
☐ D ZZZZ

Attachment No. 1

Questions 15

Annex

031-038.gif



Attachment No. 2

Questions 17

figure 84

atp_84.jpg

OPERATING CONDITIONS	H-1	H-2	H-3	H-4	H-5
ALTITUDE	24,000	17,000	8,000	18,000	22,000
WEIGHT (X1000)	195	185	155	135	175
ENGINES OPERATING	3	3	3	3	3
HOLDING TIME (MIN)	15	30	45	25	35

Figure 84. B-727 – Holding © ASA

Attachment No. 3

Questions 17

figure 85

atp_85.jpg

EPR IAS - KTS FF PER ENG - LB/HR			HOLDING							B-727	
PRESSURE ALTITUDE FT	GROSS WEIGHT - 1000 LB										
	200	190	180	170	160	150	140	130	120		
25000	1.85 268 3600	1.81 261 3400	1.77 253 3210	1.73 246 3030	1.69 238 2860	1.64 230 2680	1.60 222 2510	1.55 213 2340	1.51 205 2180		
20000	1.69 265 3630	1.66 258 3450	1.62 251 3280	1.59 244 3110	1.55 236 2940	1.51 228 2770	1.48 220 2600	1.44 212 2440	1.40 204 2270		
15000	1.56 263 3670	1.53 256 3500	1.50 249 3340	1.47 242 3170	1.44 235 3000	1.41 227 2850	1.38 219 2680	1.35 211 2520	1.32 203 2350		
10000	1.45 262 3800	1.43 255 3640	1.40 248 3460	1.38 241 3310	1.35 234 3140	1.33 226 2970	1.30 218 2810	1.28 210 2640	1.25 202 2480		
5000	1.36 260 3890	1.34 254 3720	1.32 247 3550	1.30 240 3380	1.28 233 3220	1.26 225 3060	1.24 218 2890	1.22 210 2730	1.20 201 2560		

Figure 85. B-727 – Holding Performance Chart © ASA

Comment No.2 for question 17:

not ppl material at all

Comment No.3 for question 21:

cpl and I.R work

Comment No.4 for question 40:

cpl work